

Tutorial Sheet 5

1 a)

$$\begin{array}{r} x^2 + 1 \overline{) -x^2 + 3x + 5} \\ \underline{-x^2 + 0x - 1} \\ 3x + 6 \end{array}$$

$$(-1)(x^2 + 1) + 3x + 6$$

c)

$$\begin{array}{r} x + 30 \overline{) 2x^3 + 6x^2 - x + 3} \\ \underline{-2x^3 + 0x^2 - 2x} \\ 6x^2 + x + 3 \\ \underline{-6x^2 + 0x - 6} \\ x + 9 \end{array}$$

$$(x + 3)(2x^2 - 2) + (x + 9)$$

d)

$$\begin{array}{r} x^2 - 2x + 3 \overline{) -2x^4 + 3x^3 + 1} \\ \underline{-2x^4 + 4x + 1} \\ -2x^4 + 0x^3 + 3x^2 + 0x + 1 \\ \underline{-2x^4 - 4x^3 - 6x^2} \\ 4x^3 + 9x^2 + 0x \\ \underline{4x^3 + 8x^2 + 12x} \\ x^2 - 12x + 1 \\ \underline{-x^2 + 2x + 3} \\ -14x - 2 \end{array}$$

$$(2x^2 + 4x + 1)(x^2 + 2x + 3) + (-14x - 2)$$

Q2 a)

1	-7	-4	28
3	3	-12	-48
1	-4	-16	-20

$$(x^2 - 4x - 16)(x - 3) + (-20)$$

b)

$$\begin{array}{r} 5x^2 + 6x + 46 \overline{) 5x^3 - 19x^2 + 16x - 4} \\ \underline{5x^3 - 25x^2} \\ +6x^2 + 16x \\ \underline{-6x^2 - 30x} \\ 46x - 4 \\ \underline{-46x + 230} \\ -234 \end{array}$$

$$(5x^2 + 6x + 46)(x - 5) + (-234)$$

(b) $5x + 10 = 0$

$5x = -10$

$x = -2$

	5	-6	-28	-2
-2		-10	32	-8
	5	-16	4	-10

$\therefore (5x^2 - 16x + 4)(5x + 10) + (-10)$

(c)

	1	-10	-31	-30
11		11	11	220
	1	-1	20	190

$\therefore (x^2 + x + 20)(x - 11) + (190)$

(e)

	6	+1	-21	-10
$\frac{1}{2}$		-3	1	-10
	6	-2	20	-20

$\therefore (6x^2 - 2x + 20)(2x + 1) + (-20)$

Q 30 $4x^3 - 8x^2 - x + 2$

B. $(2x - 1)$ is a factor

	4	-8	-1	2
$\frac{1}{2}$		2	-3	-2
	4	-6	-4	0

$4x^2 - 6x - 4$ Factorising gives

$4x^2 - 6x - 4$

$4x^2 - 8x + 2x - 4$

$4x^2(x - 2) + 2(x - 2)$

$(4x + 2)(x - 2)$

$\therefore (2x - 1)(4x + 2)(x - 2)$

(b) $2x^3 + 3x^2 - 3x - 2$
 $(x - 1)$ is a factor

	2	3	-3	-2
1		2	5	2
	2	5	2	0

$2x^2 + 5x + 2$

$2x^2 + 4x + x + 2$

$2x(x + 2) + 1(x + 2)$

$(2x + 1)(x + 2)$

$\therefore (x - 1)(2x + 1)(x + 2)$

(c) $x^3 - 2x^2 + 2x - 4$

$x - 2$ is a factor

	1	-2	2	-4
2		2	0	4
	1	0	2	0

$x^2 + 0x + 2$

$x^2 + 2$

$\therefore (x - 2)(x^2 + 2)$

$$d) x^3 - 1 = x^3 + 0x^2 + 0x - 1$$

$$(x^2 + x + 1)(x - 1)$$

showing steps.

$(x - 1)$ is a factor

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & -1 \\ -1 & & 1 & 1 & 1 \\ \hline & 1 & 1 & 1 & 0 \end{array}$$

$$(x^2 + x + 1)(x - 1)$$

$$e) x^3 + 1 = x^3 + 0x^2 + 0x + 1$$

$(x + 1)$ is a factor

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & 1 \\ -1 & & -1 & 1 & -1 \\ \hline & 1 & -1 & 1 & 0 \end{array}$$

$$(x^2 - x + 1)(x + 1)$$

$$f) x^4 - 6x^3 - 11x^2 + 24x - 21$$

Can't be factorised further.

$$Q4a) x^3 + x^2 - x - 1 = 0$$

$(x - 1)$ is a factor

$$\begin{array}{r|rrrr} & 1 & 1 & -1 & -1 \\ 1 & & 1 & 2 & 1 \\ \hline & 1 & 2 & 1 & 0 \end{array}$$

$$x^2 + 2x + 1$$

$$x^2 + 2x + 1$$

$$x^2 + x + x + 1$$

$$x(x + 1) + 1(x + 1)$$

$$(x + 1)(x + 1)$$

$$(x + 1)(x + 1)(x - 1) = 0$$

$$\therefore x = -1, 1$$

$$b) 2x^3 + 3x^2 - 3x - 2$$

$(x - 1)$ is a factor

$$\begin{array}{r|rrrr} & 2 & 3 & -3 & -2 \\ 1 & & 2 & 5 & 2 \\ \hline & 2 & 5 & 2 & 0 \end{array}$$

$$2x^2 + 5x + 2$$

$$2x^2 + 4x + x + 2$$

$$2x(x + 2) + 1(x + 2)$$

$$(2x + 1)(x + 2)$$

$$(2x + 1)(x + 2)(x - 1) = 0$$

$$\therefore x = -\frac{1}{2}, -2 \text{ \& } 1$$

$$c) x^3 + 6x^2 + 5x - 2$$

could not factor further

$$d) x^2(2x + 1) = 13x - 6$$

$$2x^3 + x^2 = 13x - 6$$

$$2x^3 + x^2 - 13x + 6 = 0$$

$(x + 3)$ is a factor

$$\begin{array}{r|rrrr} & 2 & 1 & -13 & 6 \\ -3 & & -6 & 15 & -6 \\ \hline & 2 & -5 & 2 & 0 \end{array}$$

$$2x^2 - 5x + 2$$

$$2x^2 - 4x - x + 2$$

$$2x(x-2) - 1(x-2)$$

$$(2x-1)(x-2)$$

$$(x+3)(2x-1)(x-2)=0$$

$$x = -3, \frac{1}{2} \text{ \& \;} 2$$

$$\textcircled{a} \quad x^4 - 6x^3 - 11x^2 + 24x + 28 = 0$$

$x+1$ is a factor

	1	-6	-11	24	28
-1		-1	7	-4	-28
	1	-7	-4	28	0

$$x^3 - 7x^2 - 4x + 28$$

$x-2$ is a factor

	1	-7	-4	28
2		2	-10	-28
	1	-5	-14	0

$$x^2 - 5x - 14$$

$$x^2 - 7x + 2x - 14$$

$$x(x-7) + 2(x-7)$$

$$(x-7)(x+2)$$

$$(x+2)(x+1)(x-2)(x-7)=0$$

$$\therefore x = -2, -1, 2 \text{ \& \;} 7$$

$$\textcircled{a} \quad x^3 - 8 = 0$$

$$x^3 + 0x^2 + 0x - 8 = 0$$

$(x-2)$ is a factor

$$x-2=0$$

$$x=2$$

	1	0	0	-8
2		2	4	8
	1	2	4	0

$$x^2 + 2x + 4$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{4 - 16}}{2}$$

$$= \frac{-2 \pm \sqrt{-12}}{2}$$

$$x = \frac{-2 + \sqrt{-12}}{2} \text{ or } x = \frac{-2 - \sqrt{-12}}{2}$$

$$\text{or } x = 2$$

$$\textcircled{a} \quad f(x) = 4x^3 - 7x - 3$$

$$\textcircled{a} \quad x=2$$

$$f(2) = 4(2)^3 - 7(2) - 3$$

$$= 4(8) - 14 - 3$$

$$= 32 - 14 - 3$$

$$= 15$$

$\therefore 15$ is the remainder when $f(x)$ is divided by $x-2$.

Q 50 $2x+1=0$

$x = -1/2$

$f(-1/2) = 4(-1/2)^3 - 7(-1/2) - 3$

$= \frac{-4}{8} + \frac{7}{2} - 3$

$= \frac{-1/2 + 7/2 - 3}{2}$

$= 0$

$= 0$

hence shown.

© $4x^3 - 7x - 3 = 0$

$(2x+1)$ is a factor.

	$4x^3 - 7x - 3$	
$-1/2$	$-2x + 1$	3
	$4x^3 - 2x - 6$	0

$4x^2 - 2x - 6 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{2 \pm \sqrt{100}}{8}$

$= \frac{2 \pm 10}{8}$

$= 12/8 \text{ or } -8/8$

$= 3/2 \text{ or } -1$

$\therefore x = -1, -1/2 \text{ and } 3/2$

Q6. a) $f(x) = ax^3 + bx^2 - 3x - 7$

$x = 1$

$a(1)^3 + b(1)^2 - 3(1) - 7 = 0$

$a + b = 10 \dots\dots (i)$

$x = -1$

$a(-1)^3 + b(-1)^2 - 3(-1) - 7 = 0$

$-a + b = 4$

$a + b = -4 \dots\dots (ii)$

$a = 10 - b$

$10 - b + (-b) = -4$

~~Return question 4 and 5~~

⑧ ~~Return question 4 and 5~~

~~Return question 4 and 5~~

$10 - b - b = -4$

$-2b = -14$

$b = 7$

$a + b = 10$

$a + 7 = 10$

$a = 10 - 7$

$a = 3$

⑩ $f(x) = x^3 + ax^2 + bx + 6$

$x = -1$

$(-1)^3 + a(-1)^2 + b(-1) + 6 = 20$

$-1 + a - b + 6 = 20$

$a - b = 15 \dots\dots (i)$

$x = 2$

$(2)^3 + a(2)^2 + b(2) + 6 = 8$

$8 + 4a + 2b + 6 = 8$

$$4a + 2b = -6 \quad \text{--- (ii)}$$

$$a = 15 + b$$

$$4(15 + b) + 2(b) = -6$$

$$60 + 4b + 2b = -6$$

$$6b = -66$$

$$b = -11$$

$$a = 15 - 11$$

$$a = 4$$

$$\therefore a = 4 \text{ \& } b = -11$$

Q7 a) $f(x) = x^3 - 13x + 12$

y-intercept, $x = 0$

$$y = 12$$

$(x - 1)$ is a factor

1	0	-13	12
1	1	1	-12
1	1	-12	0

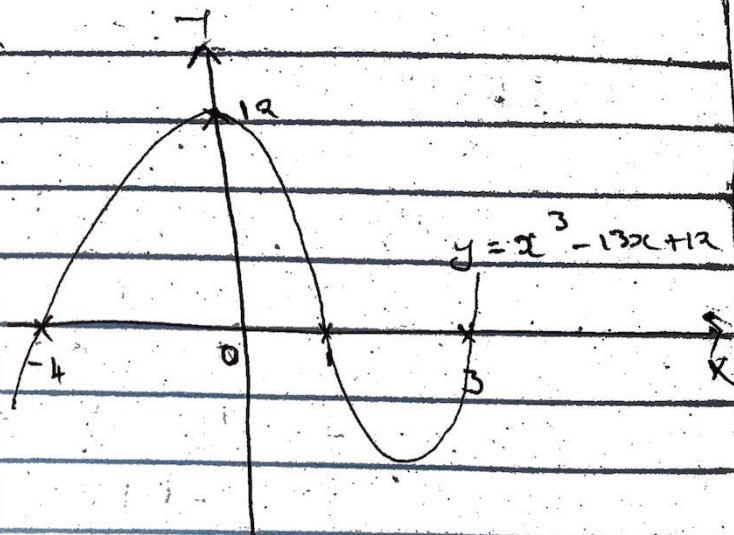
$$x^2 + x - 12$$

$$x^2 + 4x - 3x - 12$$

$$x(x + 4) - 3(x + 4)$$

$$(x - 3)(x + 4)$$

$$x = -4, 1, 3$$



b) $f(x) = 15 + 5x - 3x^2 - x^3$

y-intercept, when $x = 0$

$$y = 15 - 5(0) - 3(0)^2 - (0)^3$$

$$y = 15$$

$$-x^3 - 3x^2 + 5x + 15 = 0$$

$x + 3$ is a factor

-1	-3	5	15
-3	3	0	-15
-1	0	5	0

$$-x^2 + 0x + 5$$

$$-x^2 + 5$$

$$(x + 3)(-x^2 + 5) = 0$$

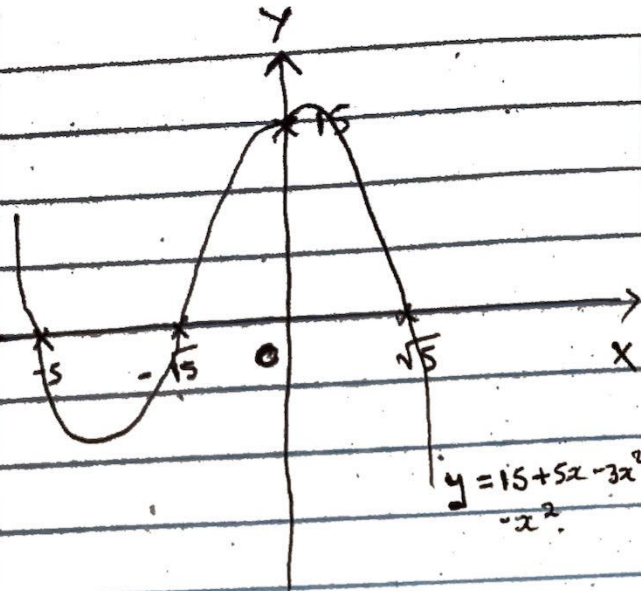
$$x = -3, \quad -x^2 + 5 = 0$$

$$\sqrt{x^2} = \sqrt{5}$$

$$x = \pm\sqrt{5}$$

x intercept

$$= -3, -\sqrt{5} \text{ \& } \sqrt{5}$$



$$\textcircled{1} 4x^3 - 12x^2 + 5x + 6 = 0$$

y - intercept, when $x = 0$

$$y = 6$$

x intercepts

$$4x^3 - 12x^2 + 5x + 6 = 0$$

$x + 2$ is a factor

	4	-12	5	6
2		8	-8	-6
	4	-4	-3	0

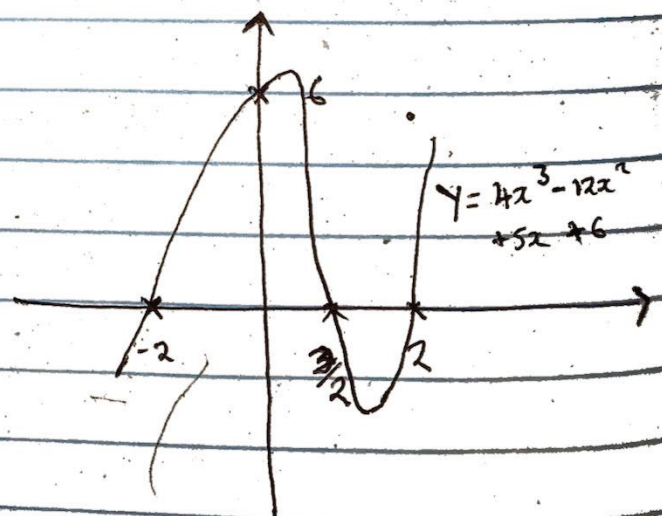
$$4x^2 - 4x - 3 = 0$$

$$4 \pm \sqrt{16 + 48}$$

$$x = 1 \text{ or } x = -1/2$$

$$\therefore x = -1/2, 3/2 \text{ \& } 2$$

$$x = -1/2, 3/2 \text{ \& } 2$$



$$\textcircled{2} f(x) = x^3 - 4x^2 + x + 6$$

$x + 1$ is a factor.

	1	-4	1	6
-1		-1	5	-6
	1	-5	6	0

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0$$

$$x^2 - 3x - 2x + 6 = 0$$

$$x(x-3) - 2(x-3) = 0$$

$$(x-2)(x-3) = 0$$

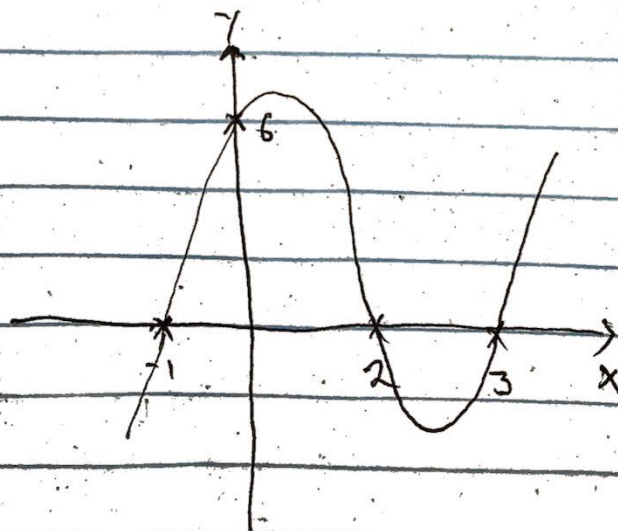
$$x = 2, x = 3$$

x - intercepts

$$x = -1, 2, 3$$

y - intercept

$$y = 6$$



$$(e) \quad x^3 - 6x - 5 = 0$$

y-intercept, when $x = 0$

$$y = -5$$

$$(x+1)(x^2-x-5)$$

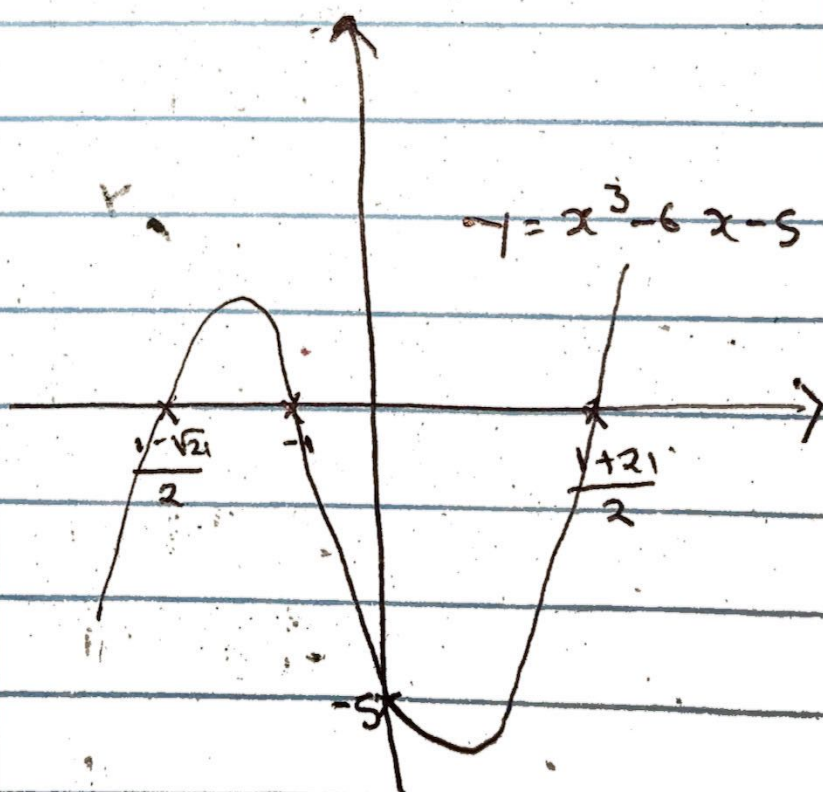
using the zero factor principle

$$x+1=0$$

$$x = -1$$

$$x^2 - x - 5 = 0$$

$$x = \frac{1 + \sqrt{21}}{2}, \quad x = \frac{1 - \sqrt{21}}{2}$$



Q7

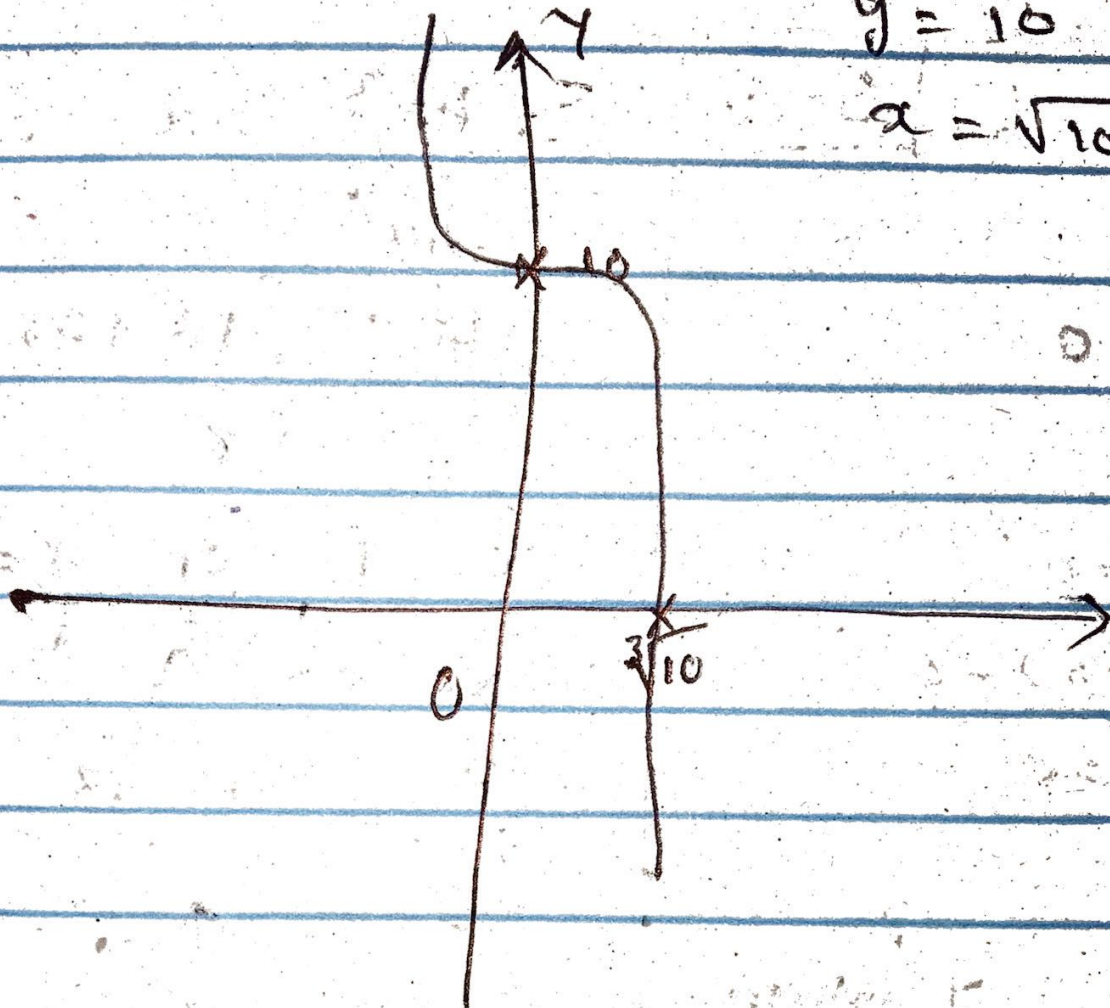
Q

Factorizing

$$-x^3 + 10 = 0$$

$$y = 10$$

$$x = \sqrt{10}$$



Q 8. (a) $f(x) = \frac{2-x}{x+4}$

$D \{x: x \neq -4, x \in \mathbb{R}\}$

Range.

$y = \frac{2-x}{x+4}$

$xy + 4y = 2 - x$

$xy + x = 2 - 4y$

$x(y+1) = 2-4y$

$x = \frac{2-4y}{y+1}$

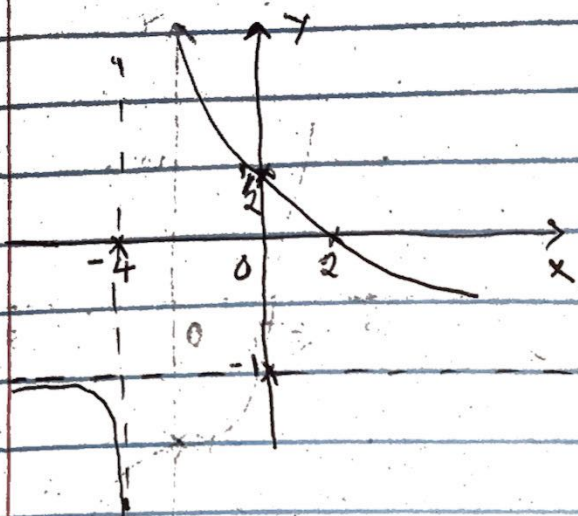
$\therefore f^{-1}(x) = \frac{2-4x}{x+1}$

Range $\{y: y \neq -1, y \in \mathbb{R}\}$

$$V.A = -4$$

$$H.A = -1$$

Cut y at $\frac{1}{2}$ & x at 2



$$\textcircled{b} \quad \frac{3x+6}{x-1}$$

Domain of $x: x \neq 1, x \in \mathbb{R}$

$$y = \frac{3x+6}{x-1}$$

$$xy - y = 3x + 6$$

$$xy - 3x = 6 + y$$

$$x(y-3) = 6+y$$

$$x = \frac{6+y}{y-3}$$

Range of $y: y \neq 3, y \in \mathbb{R}$

$$V.A = 1$$

$$H.A = 3$$

Cut at:

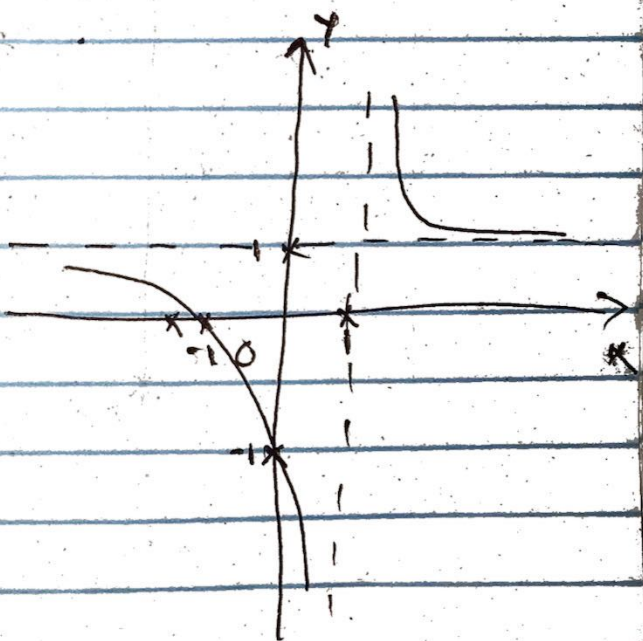
$$y = -6$$

$$x = -2$$

$$\textcircled{c} \quad \frac{x+1}{x-1}$$

Domain of $x: x \neq 1, x \in \mathbb{R}$

$$V.A = 1 \quad H.A = 1$$



Range

$$xy - y = x + 1$$

$$x(y-1) = 1+y$$

$$x = \frac{1+y}{y-1}$$

Range of $y: y \neq 1, y \in \mathbb{R}$

② $\frac{2x+5}{x-1}$

Domain $\{x: x \neq 1, x \in \mathbb{R}\}$

$$xy - y = 2x + 5$$

$$xy - 2x = 5 - y$$

$$x(y-2) = 5+y$$

$$x = \frac{5+y}{y-2}$$

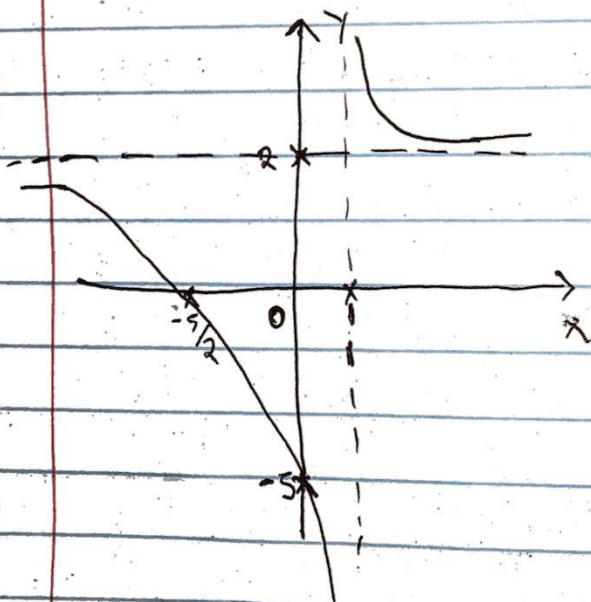
Range $\{y: y \neq 2, y \in \mathbb{R}\}$

$$V.A = 1 \quad H.A = 2$$

Cut at;

$$y = -5$$

$$x = -5/2$$



① 1

② $\frac{1}{(x+5)(x+4)}$

$$\Rightarrow \frac{A}{x+5} + \frac{B}{x+4}$$

$$\Rightarrow 1 = A(x+4) + B(x+5)$$

When $x = -4$

$$1 = B$$

$$1 = A(x+4) + B(x+5)$$

when $x = -5$

$$1 = -A$$

$$A = -1$$

Thus;

$$\frac{1}{(x+5)(x+4)} = \frac{-1}{x+5} + \frac{1}{x+4}$$

③ $\frac{x+7}{x^2+3x+2}$

Factorising the denominator.

$$x^2 + 2x + x + 2$$

$$x(x+2) + 1(x+2)$$

$$(x+1)(x+2)$$

$$\frac{x+7}{(x+1)(x+2)}$$

$$= \frac{A}{x+1} + \frac{B}{x+2}$$

$$\Rightarrow x+7 = A(x+2) + B(x+1)$$

when $x = -2$

$$(-2)+7 = A(-2+2) + B(-2+1)$$

$$5 = -B$$

$$B = -5$$

when $x = -1$

$$(-1)+7 = A(-1+2) + B(-1+1)$$

$$6 = A$$

Thus;

$$\frac{x+7}{x^2+3x+2} = \frac{6}{x+1} - \frac{5}{x+2}$$

⑥ $2x^2 - 9x - 35$
 $x^3 + 2x^2 - 5x - 6$
 $(-1)^3 + 2(-1)^2 - 5(-1) - 6$
 $-1 + 2 + 5 - 6 = 0$

$\therefore x+1$ is a factor

$x+1$	1	2	-5	-6
		-1	-1	6
	1	1	-6	0

$$x^2 + x - 6$$

$$x^2 + 3x - 2x - 6$$

$$x(x+3) - 2(x+3)$$

$$(x-2)(x+3)$$

$$\frac{2x^2 - 9x - 35}{(x+1)(x-2)(x+3)} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+3}$$

$$2x^2 - 9x - 35 = A(x-2)(x+3) + B(x+1)(x+3) + C(x+1)(x-2)$$

when $x = 2$

$$2(2)^2 - 9(2) - 35 = 15B$$

$$-45 = 15B$$

$$B = -3$$

when $x = -1$

$$2(-1)^2 - 9(-1) - 35 = -6A$$

$$-24 = -6A$$

$$A = 4$$

when $x = -3$

$$2(-3)^2 - 9(-3) - 35 = 10C$$

$$10 = 10C$$

$$C = 1$$

$$\frac{2x^2 - 9x - 35}{x^3 + 2x^2 - 5x - 6} = \frac{4}{x+1} - \frac{3}{x-2} + \frac{1}{x+3}$$

$$d \quad \frac{x^2 - 1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

$$x^2 - 1 = A(x^2 + 1) + (Bx + C)(x)$$

$$x^2 - 1 = Ax^2 + A + Bx^2 + Cx$$

$$A = -1$$

$$(A + B)x^2 = x^2$$

$$A + B = 1$$

$$-1 + B = 1$$

$$B = 2$$

$$Cx = 0$$

$$C = 0$$

$$\frac{x^2 - 1}{x(x^2 + 1)} = \frac{-1}{x} + \frac{2x}{x^2 + 1}$$

$$\frac{x-3}{x^3+3x}$$

Taking the denominator
 $x(x^2+3)$

$$\frac{x-3}{x(x^2+3)} = \frac{A}{x} + \frac{Bx+C}{x^2+3}$$

$$x-3 = A(x^2+3) + (Bx+C)(x)$$

$$x-3 = Ax^2 + 3A + Bx^2 + Cx$$

$$(A+B)x^2 = 0$$

$$-1 + B = 0$$

$$B = 1$$

$$Cx = x$$

$$C = 1$$

$$3A = -3$$

$$A = -1$$

$$\therefore \frac{x-3}{x^3+3x} = \frac{-1}{x} + \frac{x+1}{x^2+3}$$

$$h \quad \frac{4x^3 + 10x + 4}{x(2x+1)} = \frac{4x^3 + 10x + 4}{2x^2 + x}$$

$$\begin{array}{r} 2x - 1 \\ 2x^2 + x \overline{) 4x^3 + 10x + 4} \\ \underline{4x^3 + 2x^2} \\ -2x^2 + 10x + 4 \\ \underline{-2x^2 - x} \\ 11x + 4 \end{array}$$

$$2x - 1 + \frac{11x + 4}{2x^2 + x}$$

$$2x - 1 + \frac{11x + 4}{x(2x + 1)}$$

$$\frac{2x+1}{x(2x+1)} + \frac{11x+4}{x(2x+1)} = \frac{A}{x} + \frac{B}{2x+1}$$

$$11x+4 = A(2x+1) + Bx$$

$$\text{when } x=0$$

$$4 = A$$

$$11x+4 = A(2x+1) + Bx$$

$$\text{when } x = -\frac{1}{2}$$

$$-\frac{11}{2} + 4 = A(2(-\frac{1}{2}) + 1) + B(-\frac{1}{2})$$

$$-\frac{3}{2} = -\frac{B}{2}$$

$$B = 3$$

$$\frac{4x^3 + 10x + 4}{2(2x+1)} = 2x-1 + \frac{4}{x} + \frac{3}{2x+1}$$

$$(i) \frac{2x^4 + 3x^2 + 1}{x^2 + 3x + 2}$$

$$\begin{array}{r} \Rightarrow \frac{2x^4 + 3x^2 + 1}{x^2 + 3x + 2} \begin{array}{r} 2x^2 - 6x + 17 \\ \hline 2x^4 + 3x^2 + 1 \\ 2x^4 + 6x^3 + 4x^2 \\ \hline -6x^3 - x^2 + 1 \\ -6x^3 - 18x^2 - 12x \\ \hline 17x^2 + 13x + 1 \\ -17x^2 + 51x + 34 \\ \hline -39x - 33 \end{array} \end{array}$$

$$2x^2 - 6x + 17 + \frac{-39x - 33}{x^2 + 3x + 2}$$

$$\frac{2x^2 - 6x + 17}{(x+1)(x+2)} - \frac{39x - 33}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$-39x - 33 = A(x+2) + B(x+1)$$

$$\text{when } x = -1$$

$$6 = A$$

$$-39x - 33 = A(x+2) + B(x+1)$$

$$\text{when } x = -2$$

$$78 - 33 = A(-2+2) + B(-2+1)$$

$$45 = -B$$

$$B = -45$$

$$\therefore \frac{2x^4 + 3x^2 + 1}{x^2 + 3x + 2} = 2x^2 + 6x + 17 + \frac{6}{x+1} - \frac{45}{x+2}$$

W

$$\frac{4x}{x^3 + 4x^2 + 5x + 2}$$

; Taking the denominator.

$$x^3 + 4x^2 + 5x + 2, \quad x+1 \text{ is a factor.}$$

	1	4	5	2
-1		-1	-3	-2
	1	3	2	0

$$x^2 + 3x + 2$$

Factorising gives

$$(x+1)(x+2)$$

$$\frac{4x}{(x+1)(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+1} + \frac{C}{x+2}$$

$$4x = A(x+1)(x+2) + B(x+1)(x+2) + C(x+1)(x+1)$$

Expanding gives.

$$4x = (A + B + C)x^2 + (3A + 3B + 2C)x + 2A + 3B + C$$

$$A + B + C = 0$$

$$3A + 3B + 2C = 4$$

$$2A + 3B + C = 0$$

Solving simultaneously

$$C = -B - A$$

$$2A + 3B - B - A = 0$$

$$3A + 3B - 2B - 2A = 4$$

$$A + 2B = 0$$

$$A = -2B$$

$$A + B = 4$$

$$A = -2(-4)$$

$$-2B + B = +4$$

$$A = 8$$

$$-B = 4$$

$$B = -4$$

$$C = 4 - 8$$

$$C = -4$$

$$\therefore \frac{4x}{x^3 + 4x^2 + 5x + 2} = \frac{8}{x+1} - \frac{4}{x+1} - \frac{4}{x+2}$$